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AWS Certified Cloud Practitioner Foundational (CLF-C02)

Interview Preparation Cheatsheet

7 Chapters • 33 Topics • Quick-Reference Notes
Cloud Concepts | Security | Services | Billing & Support

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Chapter 1 — Cloud Concepts & Value Proposition

1.1 What is Cloud Computing & Deployment Models

- Cloud computing = on-demand delivery of IT resources (compute, storage, database) over the internet with pay-as-you-go pricing — no owning/managing physical data centers.
- Deployment models: Cloud (all-in on AWS), Hybrid (cloud + on-prem connected), On-premises/Private (your own data center). Migration often starts hybrid.
- Service models: IaaS (infrastructure — EC2, you manage OS up), PaaS (platform — Beanstalk/RDS, AWS manages infra), SaaS (software — ready apps like WorkMail).

TIP IaaS = most control (you patch the OS); SaaS = least management (just use it); PaaS is the middle ground.

1.2 Six Advantages of Cloud

- 1) Trade capital expense (CapEx) for variable expense (OpEx) — pay only for what you use.
- 2) Benefit from massive economies of scale (AWS aggregates usage → lower prices).
- 3) Stop guessing capacity — scale up/down on demand instead of over-provisioning.
- 4) Increase speed and agility — spin up resources in minutes, experiment cheaply.
- 5) Stop spending money running/maintaining data centers — focus on your business.
- 6) Go global in minutes — deploy to multiple regions worldwide easily.

TIP These six are directly exam-tested — memorize them as the AWS 'value proposition'.

1.3 Cloud Economics (CapEx vs OpEx & TCO)

- CapEx = large upfront purchase of hardware; OpEx = ongoing pay-as-you-go. Cloud shifts CapEx → OpEx and improves cash flow + agility.
- Total Cost of Ownership (TCO) includes hardware, power, cooling, staff, data-center space — cloud reduces or removes many of these hidden costs.
- Elasticity = automatically add/remove resources to match demand (no waste). Agility = ability to innovate/experiment quickly at low cost.

TIP Elasticity (auto match demand) is different from scalability (ability to grow). AWS provides both.

1.4 AWS Well-Architected Framework

- Best-practice framework built on six pillars: Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, and Sustainability.
- The Well-Architected Tool reviews workloads against these pillars and highlights risks + improvements — it's free.

TIP Mnemonic for the pillars: 'Our Skilled Reliable Performance Costs Sustain' — Ops, Security, Reliability, Performance, Cost, Sustainability.

1.5 Cloud Design Principles

- Design for failure (assume components fail), decouple components (loose coupling with queues/topics), implement elasticity (scale automatically), and automate everything.
- Think parallel + scalable, use managed services to reduce operational burden, and treat servers as disposable (cattle, not pets).

TIP Loose coupling + statelessness + automation = resilient, elastic architecture. This mindset underlies every AWS exam answer.

Chapter 2 — AWS Global Infrastructure

2.1 Regions, Availability Zones & Edge Locations

- Region = geographic area (e.g. us-east-1) containing multiple isolated Availability Zones. Choose a region by: compliance/data sovereignty, latency to users, service availability, and cost.
- Availability Zone (AZ) = one or more discrete data centers with redundant power/networking; AZs in a region are physically separated but low-latency connected.
- Edge Locations (400+) are used by CloudFront, Route 53, and Global Accelerator to cache/serve content closer to users (lower latency).

TIP Deploy across ≥ 2 AZs for high availability. Regions are isolated from each other; data doesn't leave a region unless you move it.

2.2 High Availability & Fault Tolerance Basics

- High availability = minimize downtime using redundancy (multiple AZs, load balancing, auto scaling). Fault tolerance = keep operating through component failures.
- Multi-AZ deployments (RDS Multi-AZ, ASG across AZs behind an ELB) protect against a single data-center/AZ failure.

TIP Single AZ = single point of failure. Spread resources across AZs (and regions for DR) for resilience.

2.3 Ways to Interact with AWS

- Management Console (web GUI — visual, good for learning), AWS CLI (command line — scriptable/automatable), and SDKs (in your app code, many languages).
- AWS also offers CloudShell (browser-based shell with CLI pre-configured) and mobile app for monitoring.
- Infrastructure as Code: CloudFormation (templates) and CDK provision resources repeatably instead of clicking in the console.

TIP Console = manual/visual; CLI/SDK = automation; CloudFormation = repeatable infrastructure as code.

Chapter 3 — Security & Compliance

3.1 Shared Responsibility Model

- AWS is responsible for security **OF** the cloud: hardware, global infrastructure, the physical facilities, and managed-service software.
- Customer is responsible for security **IN** the cloud: their data, IAM configuration, OS patching (on EC2), network/firewall config, and encryption settings.
- Responsibility shifts by service: with EC2 (IaaS) you patch the OS; with managed services like RDS/Lambda, AWS handles more; with S3 you configure access + encryption.

WATCH AWS never manages your data classification, IAM permissions, or customer-side encryption keys — that's always your job.

3.2 IAM: Users, Groups, Roles & Policies

- IAM manages who can access what. Users = people/apps (long-term creds), Groups = collections of users, Roles = temporary credentials assumed by services/users, Policies = JSON permissions.
- IAM is free and global (not region-specific). Roles are the secure way to give AWS services (like EC2) permissions — no hard-coded keys.
- MFA (multi-factor authentication) adds a second factor (app/hardware token) — strongly recommended, especially for the root user.

TIP Policy = Allow/Deny + Action + Resource. Attach policies to groups (not individual users) for easier management. Explicit Deny always wins.

3.3 IAM Best Practices & the Root User

- Root user (created with the account) has full access — use it only for the few tasks that require it, enable MFA, and never use it for daily work or share its keys.
- Follow least privilege (grant only what's needed), rotate credentials, use roles over long-term keys, and enable MFA.
- IAM Identity Center (formerly AWS SSO) centralizes workforce sign-in across accounts/apps.

TIP First things to do on a new account: enable MFA on root, create an admin IAM user/role, and stop using root.

3.4 AWS Security Services

- GuardDuty = intelligent threat detection (analyzes logs, no agents). Inspector = automated vulnerability scanning (EC2/containers/Lambda). Macie = discovers sensitive data (PII) in S3.
- WAF = filters web traffic (blocks SQLi/XSS) at layer 7. Shield = DDoS protection (Standard free, Advanced paid). Security Hub = central view of security findings + compliance.
- Network protection: Security Groups (instance firewall, stateful) and Network ACLs (subnet firewall, stateless).

TIP GuardDuty detects threats, Inspector finds vulnerabilities, Macie classifies sensitive data, WAF filters web attacks, Shield absorbs DDoS.

3.5 Encryption & Key Management

- Encryption at rest (data on disk) and in transit (TLS/SSL over the network) protect data. Most AWS services support encryption easily.
- AWS KMS (Key Management Service) creates and controls encryption keys, integrated across S3, EBS, RDS, and more. CloudHSM = dedicated single-tenant hardware security module.

TIP KMS = managed, easy, integrated key management (default choice). CloudHSM = when you need dedicated hardware for strict compliance.

3.6 Compliance & Governance

- AWS Artifact = self-service portal for AWS compliance reports (SOC, PCI, ISO) and agreements. AWS is compliant with many global standards.
- AWS Config = tracks resource configuration + changes + compliance rules. CloudTrail = records all API calls (who did what, when) for auditing.
- AWS Audit Manager automates evidence collection for audits; AWS Compliance Center lists supported programs.

TIP Artifact = download compliance docs; Config = is it configured right; CloudTrail = who did it (audit trail).

Chapter 4 — Compute, Storage & Networking

4.1 Amazon EC2 & Purchasing Options

- EC2 = resizable virtual servers in the cloud. You pick instance type (CPU/memory/network), OS (AMI), and storage. Full control — you manage the OS.
- Pricing options: On-Demand (pay per second, no commitment), Reserved Instances (1/3-yr commit, big discount), Savings Plans (flexible commit for discount), Spot (up to 90% off, interruptible), Dedicated Hosts (physical server, compliance/BYOL).
- Instance families: general purpose, compute optimized, memory optimized, storage optimized, accelerated (GPU).

TIP On-Demand = short/unpredictable; Reserved/Savings Plans = steady long-term; Spot = flexible/fault-tolerant batch; Dedicated = compliance/licensing.

4.2 Other Compute Services

- AWS Lambda = serverless functions; run code without managing servers, pay per request + duration. Great for event-driven tasks.
- Containers: ECS (AWS container orchestration) and EKS (managed Kubernetes); Fargate runs containers serverless (no servers to manage).
- Elastic Beanstalk = deploy/manage web apps easily (PaaS). Lightsail = simple VPS with predictable pricing for small apps/websites.
- Batch = run batch jobs; Outposts = AWS hardware on-prem; App Runner = simple containerized web apps.

TIP No servers to manage → Lambda/Fargate. Simple website → Lightsail. Easy web-app deploy → Beanstalk. Full control → EC2.

4.3 Storage Services

- Amazon S3 = object storage (files/backups/static websites/data lakes), virtually unlimited, 11 9s durability, storage classes for cost (Standard, Intelligent-Tiering, Glacier for archive).
- Amazon EBS = block storage volumes attached to EC2 (like a hard drive, single AZ). Instance Store = temporary storage physically attached (lost on stop).
- Amazon EFS = shared file storage for Linux (many EC2 at once); FSx = managed file systems (Windows, Lustre). Storage Gateway = hybrid on-prem ↔ AWS storage.

TIP Objects/files/backups → S3; EC2 disk → EBS; shared Linux files → EFS; archive → S3 Glacier; hybrid → Storage Gateway.

4.4 Networking & Content Delivery

- Amazon VPC = your isolated virtual network (subnets, route tables, gateways). Security Groups (instance firewall) and NACLs (subnet firewall) control traffic.
- Route 53 = DNS + domain registration + health checks + routing policies. CloudFront = global CDN caching content at edge locations for low latency.
- Connectivity: Internet Gateway (public internet), NAT Gateway (private subnet outbound), Direct Connect (dedicated private link), VPN (encrypted over internet).

TIP VPC = your private network; Route 53 = DNS; CloudFront = speed up content globally; Direct Connect = private dedicated on-prem link.

4.5 Load Balancing & Auto Scaling

- Elastic Load Balancing (ELB) distributes traffic across multiple targets/AZs for HA. Types: ALB (HTTP/HTTPS), NLB (TCP/UDP, high performance), GWLB (appliances).
- EC2 Auto Scaling automatically adds/removes instances to match demand (min/desired/max), improving availability + cost efficiency.

TIP ELB + Auto Scaling across multiple AZs = the classic highly available, elastic web architecture.

Chapter 5 — Databases, Integration & Management

5.1 Database Services

- Amazon RDS = managed relational databases (MySQL, PostgreSQL, MariaDB, Oracle, SQL Server) — AWS handles backups, patching, HA (Multi-AZ). Aurora = AWS's high-performance MySQL/PostgreSQL-compatible DB.
- DynamoDB = serverless NoSQL key-value database, single-digit ms latency, auto-scales. ElastiCache = in-memory cache (Redis/Memcached) to speed up apps.
- Redshift = data warehouse for analytics (OLAP). Other purpose-built: Neptune (graph), DocumentDB (MongoDB-compatible), Timestream (time-series), QLDB (ledger).

TIP Relational → RDS/Aurora; NoSQL key-value → DynamoDB; cache → ElastiCache; analytics/warehouse → Redshift.

5.2 Migration & Data Transfer

- Database Migration Service (DMS) migrates databases to AWS with minimal downtime. Schema Conversion Tool (SCT) converts schema between different engines.
- DataSync = fast bulk data transfer online. Snow Family (Snowcone/Snowball) = physical devices to move large data offline when the network is too slow.
- Migration Hub tracks migrations; Application Discovery Service inventories on-prem servers.

TIP Online DB move → DMS; large offline data → Snowball; ongoing file transfer → DataSync.

5.3 Application Integration

- Amazon SQS = message queue to decouple components (one service sends, another processes later). Amazon SNS = pub/sub notifications (fan-out to email/SMS/Lambda/SQS).
- EventBridge = event bus connecting apps + AWS + SaaS. Step Functions = orchestrate workflows across services.

TIP Decouple with a queue (SQS) or notify many subscribers (SNS). Decoupling makes systems resilient and scalable.

5.4 Management & Monitoring

- CloudWatch = monitoring: metrics, alarms, logs, dashboards. CloudTrail = audit log of API activity. AWS Config = resource configuration + compliance tracking.
- Trusted Advisor = recommendations across cost, performance, security, fault tolerance, and service limits. Systems Manager = operational management (patching, run commands, parameters).
- AWS Health Dashboard shows service + account health/events.

TIP CloudWatch = performance/health; CloudTrail = who did what; Config = compliance; Trusted Advisor = optimization recommendations.

5.5 Developer, Deployment & Other Services

- Developer tools: CodeCommit (git repos), CodeBuild (build), CodeDeploy (deploy), CodePipeline (CI/CD orchestration). CloudFormation = infrastructure as code.

- Analytics: Athena (SQL on S3), Glue (ETL), Kinesis (streaming), QuickSight (BI dashboards), EMR (big data).
- AI/ML: SageMaker (build/train models), plus ready-to-use AI: Rekognition (images), Comprehend (NLP), Polly (text-to-speech), Transcribe (speech-to-text), Translate, Lex (chatbots), and Amazon Q / Bedrock (generative AI).

TIP You don't need deep detail on each for CLF-C02 — know what category each service belongs to and the problem it solves.

Chapter 6 — Billing, Pricing & Support

6.1 AWS Pricing Fundamentals

- Three core pricing principles: pay-as-you-go (pay only for what you use), pay less when you reserve (commitment discounts), and pay less as you use more / as AWS grows (volume + price drops).
- AWS Free Tier: Always Free (e.g. Lambda 1M requests/mo, DynamoDB 25 GB), 12-Months Free (e.g. 750 hrs/mo t2/t3.micro EC2, 5 GB S3), and Trials (short-term).
- You generally pay for compute, storage, and outbound data transfer. Inbound data transfer is usually free; data transfer OUT to the internet costs.

TIP Data transfer INTO AWS = free; data transfer OUT to internet and across regions = charged. Same-AZ traffic is cheapest.

6.2 EC2 Pricing Models

- On-Demand (no commitment, highest per-hour), Reserved Instances (1 or 3 years, up to ~72% off), Savings Plans (flexible commitment \$/hr for discount), Spot (up to 90% off, interruptible), Dedicated Hosts (physical, licensing/compliance).

TIP Steady workloads → Savings Plans/Reserved; spiky/short → On-Demand; fault-tolerant batch → Spot; per-socket licensing → Dedicated Hosts.

6.3 Billing & Cost Management Tools

- AWS Pricing Calculator = estimate costs before deploying. AWS Cost Explorer = visualize + analyze past/forecast spending. AWS Budgets = set alerts when cost/usage exceeds a threshold.
- Cost & Usage Report (CUR) = most detailed billing data. Cost Allocation Tags group spend by project/team. Cost Anomaly Detection flags unexpected spikes.

TIP Estimate → Pricing Calculator; analyze → Cost Explorer; alert → Budgets; deep detail → Cost & Usage Report.

6.4 AWS Organizations & Consolidated Billing

- AWS Organizations manages multiple accounts centrally, with consolidated billing (one bill), combined volume discounts, and shared Savings Plans/Reserved Instances.
- Service Control Policies (SCPs) set permission guardrails across accounts. Group accounts into Organizational Units (OUs) for policy + billing structure.

TIP Consolidated billing benefits: one payment, volume-tier pricing across accounts, and shared reservations — while keeping accounts isolated.

6.5 AWS Support Plans

- Basic (free): documentation, forums, Trusted Advisor (core checks), Personal Health Dashboard, plus account/billing support.
- Developer: business-hours email support, general guidance — for experimenting/dev.
- Business: 24/7 phone/email/chat, full Trusted Advisor, production workloads, API access — the common production choice.

- Enterprise On-Ramp: adds a pool of Technical Account Managers (TAMs) + faster response for business-critical systems.
- Enterprise: dedicated TAM, Concierge (billing) support, fastest response (~15 min for business-critical), Infrastructure Event Management.

TIP Need 24/7 production support → Business (minimum). Need a dedicated TAM + fastest response → Enterprise.

6.6 Additional Resources & Roles

- Technical Account Manager (TAM) = your dedicated technical contact (Enterprise/On-Ramp). AWS Professional Services + Partner Network (APN) help with projects.
- AWS Marketplace = buy/deploy third-party software. AWS Knowledge Center, whitepapers, and re:Post (Q&A) are free learning resources.

TIP TAM is only in Enterprise/Enterprise On-Ramp support — a common exam distractor.

Chapter 7 — Exam Preparation

7.1 CLF-C02 Exam Overview & Domains

- 65 questions, 90 minutes, scaled score 100–1000, pass = 700. \$100 USD. Multiple choice + multiple response. Valid 3 years. No deep technical/coding required.
- Domains: (1) Cloud Concepts 24%, (2) Security & Compliance 30%, (3) Cloud Technology & Services 34%, (4) Billing, Pricing & Support 12%.
- Foundational cert — tests breadth (what services do + AWS value/security/pricing basics), not deep architecture. Good first step before Associate certs.

TIP Security & Compliance + Technology & Services = ~64% of the exam. Prioritize the Shared Responsibility Model, IAM, and core service purposes.

7.2 Key Service Selection Cheat (Keyword → Service)

- 'Serverless / no servers' → Lambda / Fargate. 'Object storage / static website' → S3. 'Virtual server' → EC2. 'Relational DB' → RDS/Aurora. 'NoSQL' → DynamoDB.
- 'CDN / cache content globally' → CloudFront. 'DNS' → Route 53. 'Decouple' → SQS/SNS. 'Data warehouse / analytics' → Redshift. 'In-memory cache' → ElastiCache.
- 'Compliance reports' → Artifact. 'Audit API calls' → CloudTrail. 'Monitoring/alarms' → CloudWatch. 'Recommendations' → Trusted Advisor. 'Threat detection' → GuardDuty.
- 'Estimate cost' → Pricing Calculator. 'Analyze spend' → Cost Explorer. 'Cost alerts' → Budgets. 'Move a database' → DMS. 'Move huge data offline' → Snowball.

TIP Most CLF-C02 questions are 'which service does X' — matching keywords to services quickly is the highest-yield skill.

7.3 Exam Day Strategy & Next Steps

- ~1.4 min/question — answer all (no penalty for guessing), flag-and-return on tricky ones, and eliminate obviously wrong options first.
- Watch qualifiers: MOST cost-effective, LEAST operational overhead, MOST secure, highly available — they change the 'best' answer among plausible options.
- Free study resources: AWS Skill Builder, exam guide + sample questions, and the AWS Cloud Practitioner Essentials course. Do timed practice; aim 80%+.
- Next steps after CLF-C02: Solutions Architect Associate (SAA-C03), Developer Associate, or SysOps Associate; then Professional-level certs.

TIP Hands-on in the Free Tier (launch an EC2, create an S3 bucket, set up IAM/MFA) cements the concepts far better than memorizing.